

Talha Bilal

Senior Software Engineer & Technical Team Lead

Java | Spring Boot | PKI | Cryptography | Remote

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PROFESSIONAL SUMMARY

Senior engineer and technical lead with 4+ years building secure enterprise platforms on Java Spring Boot and PKI. Designs microservices for certificate lifecycle, digital signing, and HSM integration (AWS CloudHSM, Utimaco, SoftHSM2). Experienced in Spring Security, REST/OpenAPI, Bouncy Castle, PKCS#11, and leading distributed engineering teams.

CORE SKILLS

Java 21/17, Spring Boot, Spring Security, Microservices, REST, OpenAPI, PKI, PKCS#11, Bouncy Castle, IAIK, S/MIME, AWS CloudHSM, Docker, Jenkins, RabbitMQ, Redis, PostgreSQL, MySQL

EXPERIENCE

Dictalabs - Technical Team Lead (Java Spring Boot & PKI)

Aug 2024 - Present

- Lead team delivering microservices PKI and cryptographic systems.
- Architect HSM services for key management, signatures, and RFC 3161 timestamping.
- Own PKCS#11 integrations (AWS CloudHSM, SoftHSM2); improve signing reliability.

Codegic - Senior Software Engineer (Java Spring Boot & PKI)

Aug 2021 - Jul 2024

- Built CA, OCSP/CRL, and certificate lifecycle management.
- Developed S/MIME/CMS signing and Crypto-Core microservices with HSM integration.
- Implemented CI/CD, Docker, and performance tuning for production crypto services.

SELECTED PROJECTS

- PKI Server - CA lifecycle, OCSP/CRL, HSM-backed issuance
- Secure Email Signing - S/MIME/CMS with RSA & ECDSA
- Workflow Management - PDF signing, approvals, audit trails
- Timestamp Authority - RFC 3161 compliant TSA
- PKCS#11 Service - REST layer for enterprise HSMs
- Crypto-Core - Key generation, CSR signing, tenant isolation

EDUCATION

B.S. Computer Science - KFUEIT, 2021